Darmanis et al. 2017 — Glioblastoma

Brain · SmartSeq2 (TPM) · 1,091 malignant cells (4/4 samples) · embryonic-program cell-state analysis

GO / NO-GO GATES

Gate 1 Mixing + continuity GO

4 samples intermix per-neighbourhood (iLISI 1.09→2.89 of 4) on one continuous manifold (single kNN component, 0 islands); no per-patient islands.

Gate 2 Module geography GO

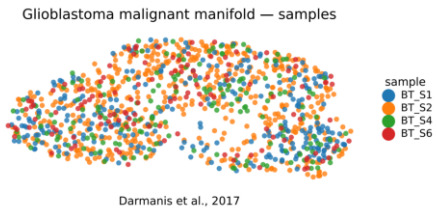
Gast/pSom co-vary (r=+0.66) and both oppose Crest (−0.97 / −0.75): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 Cross-repression NA

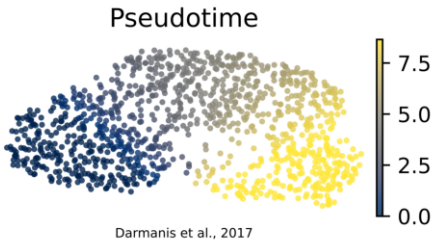
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

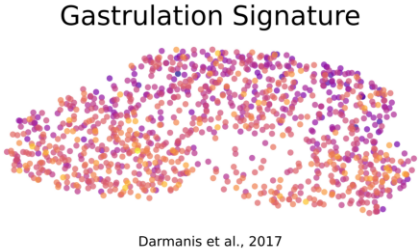
Samples (integration)



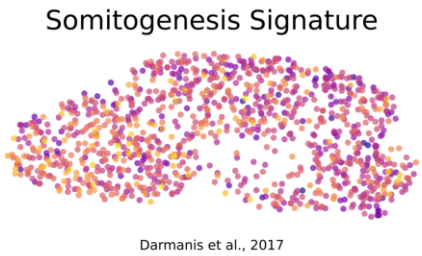
Pseudotime



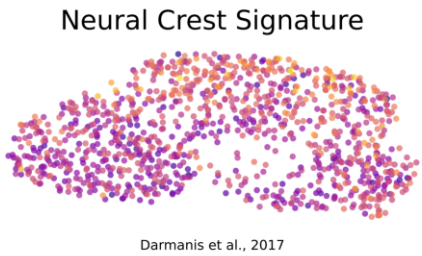
Gastrulation



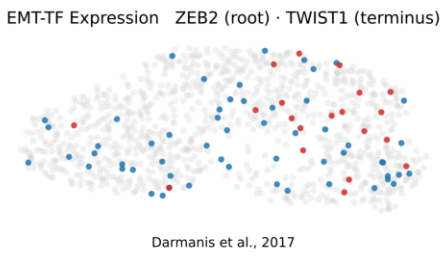
Somitogenesis



Neural Crest



EMT-TF expression



PARAMETER CHOICES & RATIONALE

Min-cell threshold:	≥100 → 4/4 samples, 1,091 cells (counts 531/268/163/129; ratios <2, no clean gap)
Expression format:	matrix pre-log-normalized (range 0.7–18.7, not linear TPM); undo loader's 2nd log1p (expm1)
Genes / embedding:	139 overdispersed (GAM); n_pcs=20 (low scree elbow rejected); Harmony θ=10
Islands removed:	0 (single connected kNN component; radius-graph = 1 component)
Geometry:	Non-branching — scFates curve (Nodes=20, λ=0.3, μ=100); 2 tips, 0 forks, fold-free (2.68)
Root:	Curve tip 0 — Gast/pSom-high, Crest-low (developmental origin; Crest is late)
EMT-TFs:	ZEB2 (51% expr, root-trending) · TWIST1 (18% expr, terminus-trending); Snail/FoxC near-absent

VERDICT

The embryonic-program pattern REPRODUCES as the non-branching case: a single continuous gastrulation/ somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Linear-pSom/Crest geometry (pSom μ=+0.44 strongest).